PAPER_513: NGC 1277 Wolfram Hypergraph Spacetime Dimension

Unified Quantum Field Framework (UQFF)

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Star Magic UQFF Framework — Session 138

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Abstract

NGC 1277 hosts one of the most overmassive black holes known relative to its host galaxy: M_BH ≈ 17 billion M■, constituting ≈14% of the galaxy's bulge mass. The Wolfram hypergraph BFS dimension estimator in SOURCE179 computes D_eff from causal graph growth rates and applies UQFF correction δD = k_PCR × (D_eff − 3). Near ultramassive BHs, spacetime curvature should produce measurable D_eff deviations above 3.

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1. Hypergraph BFS Dimension Formula

$$ D_\text{eff}(d) = \frac{\log N_\text{BFS}(d)}{\log d}, \quad N_\text{BFS}(d) \approx \min\bigl(b^d,\, N_\text{total}\bigr) $$

where b = mean branching factor (4.5 for compact/curved region), d = BFS depth.

UQFF correction: $$ \delta D = k_\text{PCR}\cdot(D_\text{eff} - 3) $$

$$ D_\text{corrected} = D_\text{eff} + \delta D $$

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2. NGC 1277 Parameters

Parameter	Value
BH mass	M_BH = $1.7\times10^{1■}$ M■
Schwarzschild radius	r_s = 50 AU
Host galaxy stellar mass	M_* ≈ 1.2×10^{11} M■
BH/bulge ratio	≈14% (vs typical 0.1%)
Distance	73 Mpc (Perseus cluster)
Velocity dispersion	σ ≈ 333 km/s

3. Computed Result

For branching factor $b = 4.5$, depth $d = 6$:

$$ N_\text{BFS}(6) = 4.5^6 \approx 8303 \Rightarrow D_\text{eff} = \frac{\log 8303}{\log 6} \approx 4.83 $$

$$ \delta D = 0.314\times(4.83 - 3) = 0.575 \Rightarrow D_\text{corrected} \approx 5.40 $$

A D_corrected ≈ 5.4 indicates the UQFF framework interprets the extreme curvature near NGC 1277's BH as requiring 2 additional effective dimensions — consistent with 26D layer gravity framework predictions.

4. Validation

- C++ term: `SOURCE179::NGC1277_HypergraphD_Term` → `NGC1277_HypergraphDimension`
- CP2 class: `NGC1277HypergraphDimCalculator` → D_eff, D_corrected, δD, k_PCR

References

- van den Bosch et al. (2012) *An overmassive black hole in the compact lenticular galaxy NGC 1277*, Nature 491, 729
- Walsh et al. (2016) *Revisiting the NGC 1277 black hole mass*, ApJ 817, 2
- Murphy, D.T. *PAPER_507: WolframFieldUnityEngine Hypergraph*